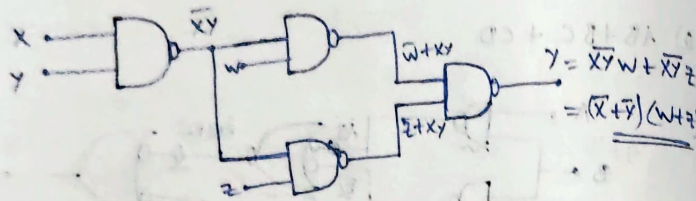


(Q) $(\bar{x} + \bar{y}) \cdot (w + z)$

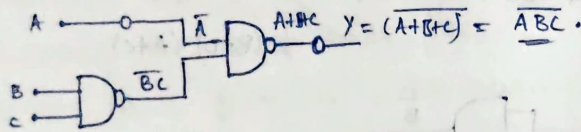
$$= (\overline{xy}) \cdot (w + z)$$

$$= \underline{\underline{\bar{x}\bar{y}w + \bar{x}\bar{y}z}}$$



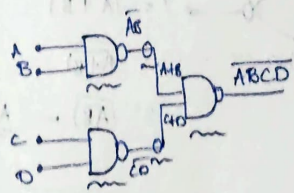
(Q) $\bar{A}\bar{B} + \bar{A}\bar{C} = \bar{A}(\bar{B} + \bar{C})$

$$= \bar{A}(\overline{B+C}) = \underline{\underline{\bar{A} \cdot \overline{BC}}}$$



END OF DAY 2 //

(Q) $\bar{A}\bar{B} + \bar{C}\bar{D} = \overline{ABCD}$



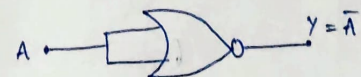
5 gates.

⇒ Minimum no. of Nor Gate :-

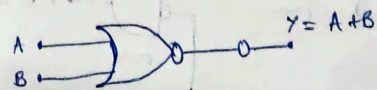
- 1 Minimize function. [Try converting it in Pos]
- 2 Implement the function using Basic gates.
- 3 Convert Basic gates into Nor gates.

	No. of Nor gates
NOT	1
OR	2
AND	3

(Q) $\bar{A}$

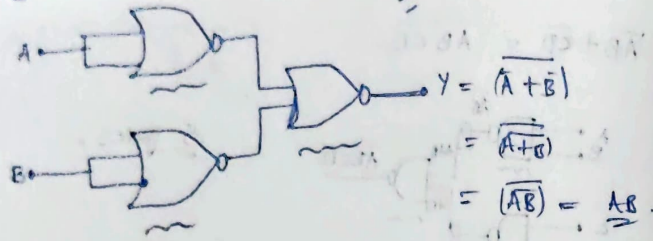


(Q) $A + B$



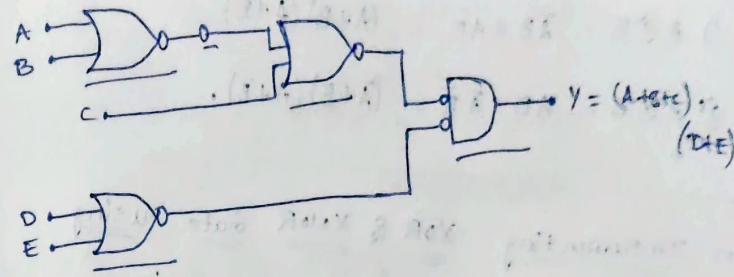
(Q) $\overline{AB}$

3



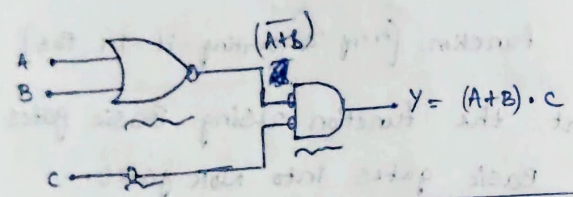
(Q) $(A+B+C)(D+E)$

5



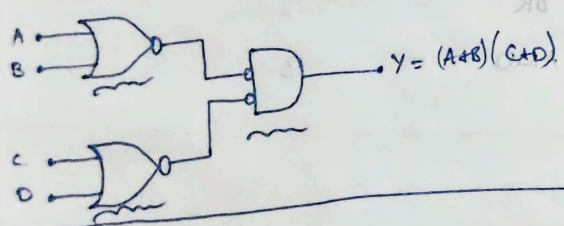
(Q) $(A+B)C$

3



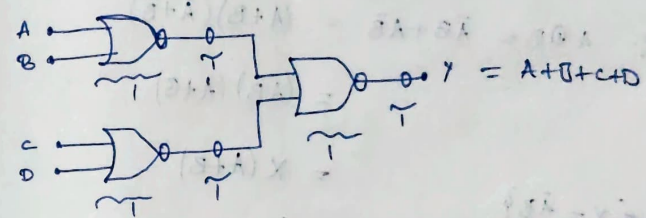
(Q) $(A+B)(C+D)$

3

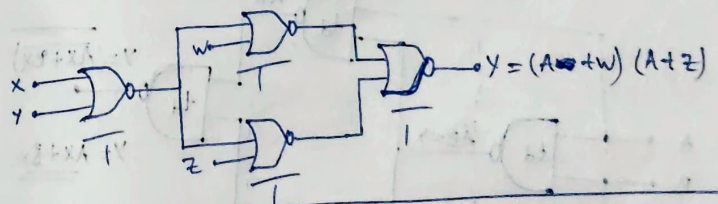


(Q) $A+B+C+D$

6

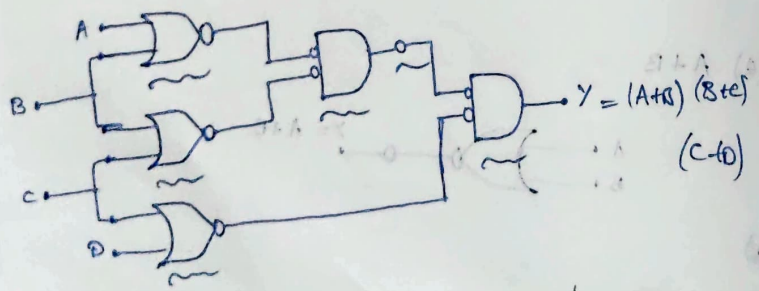


(Q) $\overline{X}Y + WZ$ (SOP) = $(\overline{X+Y}) + WZ$ $\overline{X+Y} = A$
 $= (A+W)(A+Z)$ (POS) 4

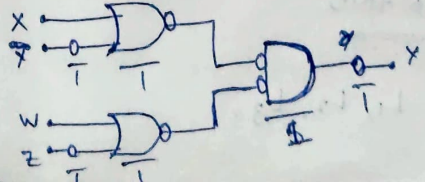


(Q) $(A+B)(B+C)(C+D)$

6



(Q) $\overline{X}Y + \overline{W}Z$ (SOP) = $\overline{(\overline{X}Y + \overline{W}Z)}$ = $\overline{(\overline{X} \cdot Y)} \cdot \overline{(\overline{W} \cdot Z)}$
 $= \overline{(X+\overline{Y})} \cdot \overline{(W+\overline{Z})}$ (POS)



6

(*) Important Points:-

$$① A \oplus B = \bar{A}B + A\bar{B} = (A+B)(\bar{A}+\bar{B})$$

$$② A \odot B = AB + \bar{A}\bar{B} = (\bar{A}+B)(A+\bar{B})$$

⇒ Implementing XOR & XNOR Gate using NAND:-

NAND:-

⇒ NAND:-

$$① \text{ XOR:- } A \oplus B = \bar{A}B + A\bar{B} = (A+B)(\bar{A}+\bar{B})$$

$$= (\bar{A}\bar{B})(A+B)$$

$$= X(A+B)$$

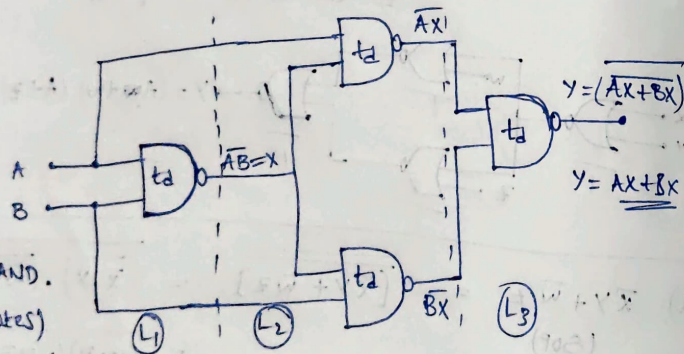
$$= \underline{AX + BX}$$

$$\{X = \bar{A}\bar{B}\}$$

①

③

3+1 = 4 gates



XOR using NAND

3-level assembly - L1, L2, L3.

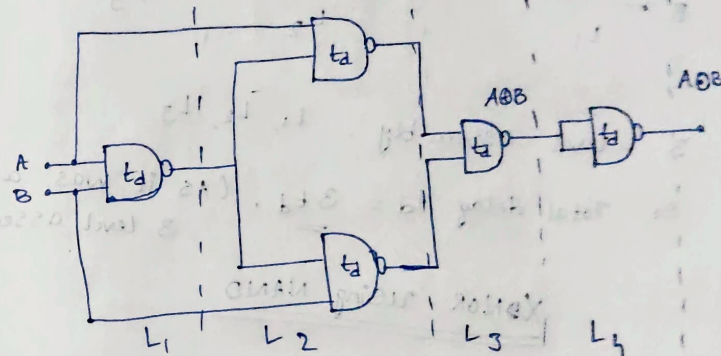
(*) Propagation delay of one NAND gate = t_d

Total Propagation delay = $3t_d$

$$② \text{ XNOR:- } A \odot B = \overline{A \oplus B} = AB + \bar{A}\bar{B}$$

~~(XOR)~~

⇒ Way - 1 :- let $\{X = \bar{A}\bar{B}\}$



⇒ Total Delay = $5t_d$ (5 NAND gates)

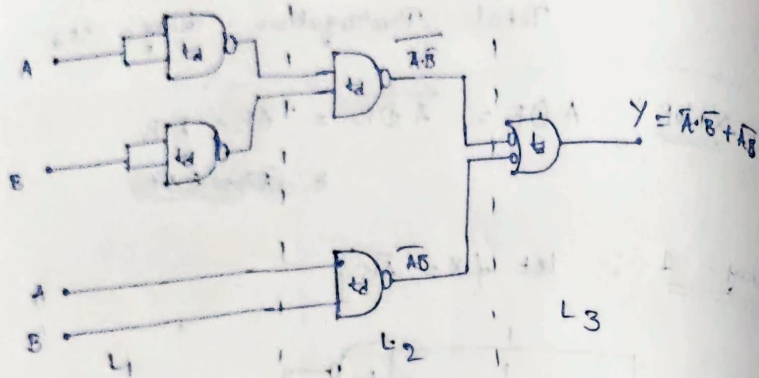
4-level Assembly - L1, L2, L3, L4.

$$\Rightarrow \text{Way-2:- } Y = A \odot B = \bar{A}\bar{B} + AB$$

$$\begin{matrix} \downarrow \downarrow & \downarrow \\ 11 & 1 \end{matrix}$$

$$3+1+1 = 5 \text{ gates}$$

But, we can get less delay.



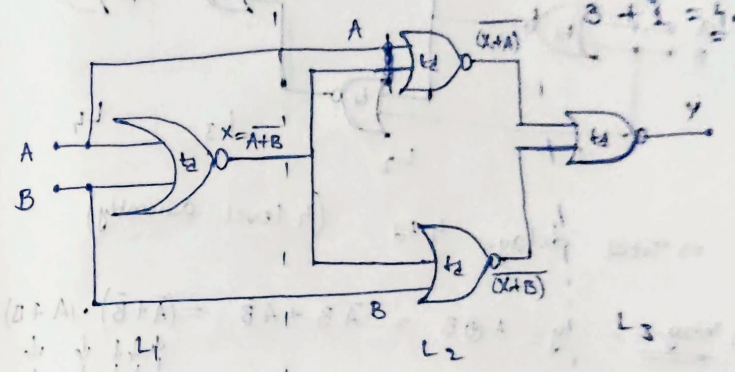
3-level assembly - L_1, L_2, L_3
 So, Total delay $t_d = 3t_d$. (As it was a 3 level assembly)

XNOR using NAND

⇒ Way 1 is used, if we want both XOR and XNOR in our design.
 ⇒ Way 2 is used, if we want only XNOR in our design.

(4) XOR & XNOR gates using NOR gate :-

XNOR
 ① ~~XOR~~ :- $Y = A \odot B = AB + \overline{A}\overline{B}$ ($Y = \overline{A \oplus B}$)
 $= AB + (\overline{A+B})$
 $= X + AB = (X+A) \cdot (X+B)$
 $\downarrow \downarrow \downarrow$
 $2 \quad 2 \quad 2$



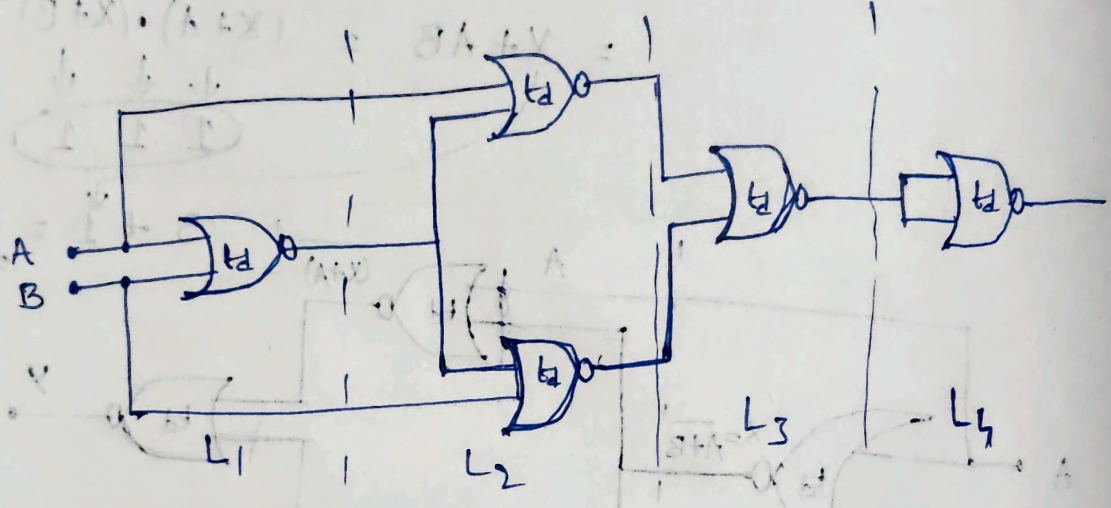
⇒ Total delay = $3t_d$.

XNOR using NOR

② XOR :- $y = A \oplus B$

$= \overline{A}B + A\overline{B}$

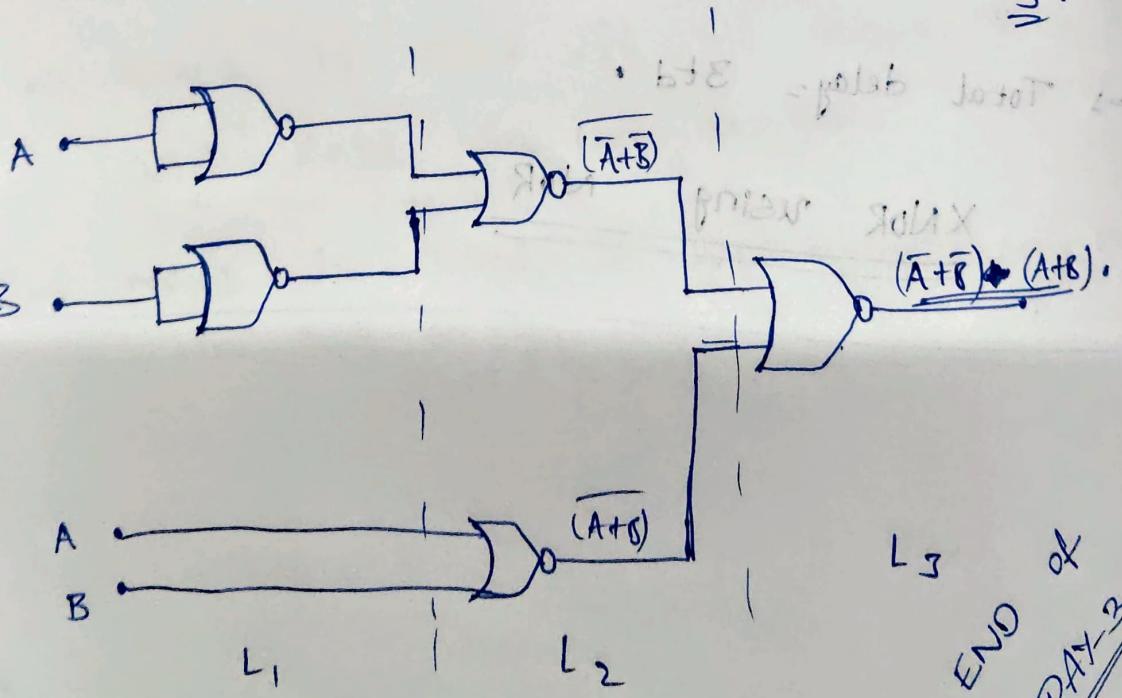
→ Way-1 :- $A \oplus B = \overline{A \odot B}$



⇒ Total delay = $4t_d$ (4 level assembly)

⇒ Way-2 :- $y = A \oplus B = \overline{A}B + A\overline{B} = (\overline{A+B}) \cdot (A+B)$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $\overline{A+B} \quad A+B$
5.



3-level assembly - L_1, L_2, L_3 .

END of DAY-3